

# InnerVoice AI: Explore the emotional patterns behind text

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# Motivation

**01**

People often don't realize **how their Facebook posts are perceived** by others.

**02**

Tone or writing style may seem **rude, emotional, or negative** unintentionally.

**03**

**Language and tone reflect personality and emotions,** shaping how others see us.

**04**

Misunderstandings occur due to **differences in tone or word choice.**

# Motivation

**05**

**No significant research or dataset** exists for Bangla Facebook text analysis.

**06**

There is a need for an **AI tool** that helps users express themselves **clearly and positively**.

# Problem Description

- ▶▶ No AI tool analyzes Bangla Facebook posts for emotion and personality.
- ▶▶ Bangla NLP research and datasets are very limited.
- ▶▶ Most tools focus on English, not Bangla slang
- ▶▶ Users can't see how their tone is perceived by others.
- ▶▶ We need an AI system that connects language, emotion, and personality.

# Scopes of the Project

- 1 Develop an AI-powered Bangla Facebook Post Analyzer using Deep Learning-based NLP.
- 2 Detect sentiment (positive, negative, neutral) and emotion (joy, anger, sadness, fear, surprise).
- 3 Predict user personality traits based on the OCEAN model.
- 

OCEAN model =

O = Openness (creative or traditional)

C = Conscientiousness (organized or careless)

E = Extraversion (social or reserved)

A = Agreeableness (kind or harsh)

N = Neuroticism (calm or emotionally unstable)

E

A

N =

# Scopes of the Project (Cont.)

4

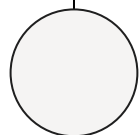
Use CNN-BiLSTM architecture with BanglaBERT / FastText embeddings for accurate text understanding.

5

Provide AI rewriting suggestions for more positive and empathetic expression.

6

Enable emotional trend tracking to observe personality and mood changes over time.



# Challenges

1. Lack of large, labeled Bangla emotional and personality datasets.
2. Ambiguity and context sensitivity in Bangla language and slang.
3. Limited pretrained Bangla NLP models for deep emotion/personality mapping.
4. Designing context-aware AI rewriting that maintains tone and intent.
5. Evaluating psychological accuracy and ensuring ethical AI suggestions.

# Registration page



## InnerVoice AI

Understand your emotions, improve your communication, and track your emotional growth over time.

AI  
Powered Analysis

24/7  
Always Available

100%  
Private & Secure



Sign In

### Create Your Account

Start your journey to better emotional awareness

Username \*

Rishad1453

Email \*

you@example.com

Password \*

.....

✓ At least 8 characters

✓ Contains lowercase letter

✓ Contains uppercase letter

✓ Contains a number

Confirm Password \*

Confirm your password

☐ I agree to the Terms of Service and Privacy Policy

Create Account →

Already have an account? [Sign in](#)

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# Login page



## InnerVoice AI

Understand your emotions, improve your communication, and track your emotional growth over time.

AI  
Powered Analysis

24/7  
Always Available

100%  
Private & Secure

[Create Account](#)

### Welcome Back

Enter your credentials to access your account

**Username \***

**Password \***

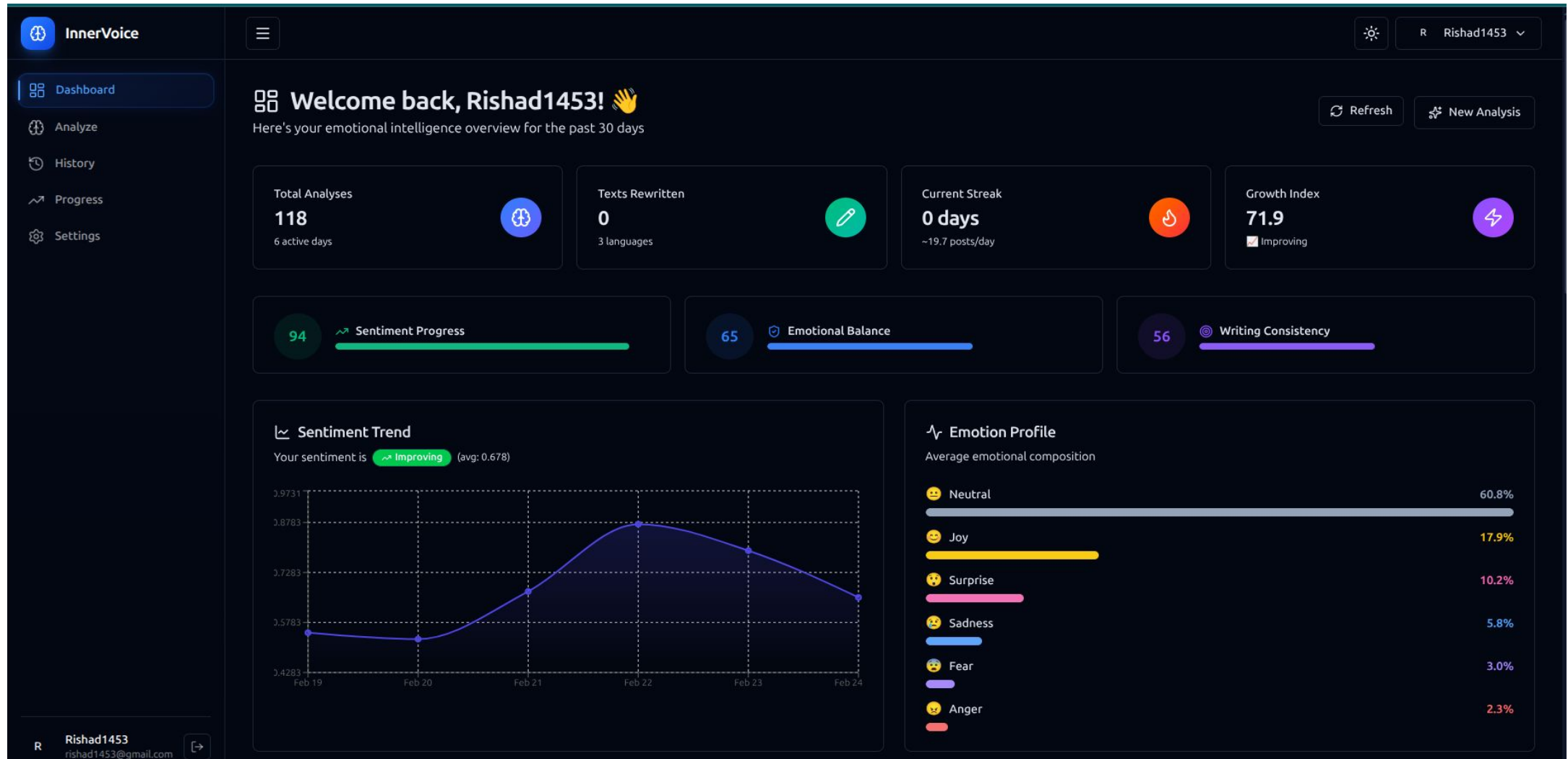
☐ Remember me [Forgot password?](#)

Don't have an account? [Create one](#)

**Demo Credentials**  
Username: demo  
Password: Demo1234

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# Dashboard page



# Analysis page

 InnerVoice

Dashboard

Analyze

History

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Settings



Text Analysis

Analyze your text for emotional insights and personality traits

 Enter Your Text

Type or paste the text you want to analyze

Enter your text here... (e.g., 'I'm feeling excited about the new project, but also a bit nervous about the deadline.')

0 / 5000 charactersMin: 10 characters

Analyze Text

Rewrite Suggestion

 Analysis Results

Emotional insights from your text

Sentiment Analysis

Emotion Analysis

Personality Insights



Your analysis results will appear here

Enter text and click "Analyze Text" to get started

# Rewriting page

R Rishad1453

[← Back to Analyze](#)



## AI Text Rewriter

Transform your text based on emotional and personality analysis



### Original Text

Emotional words are highlighted based on your analysis

hello how are u

CopyEdit



### Custom Rewrite Instruction

Describe how you want to transform the text. Works in English or Bangla/Banglish.

e.g. "increase sadness" • "make it more formal" • "sadness komao" • "Bangla te lekho"

0/300 — press Ctrl+Enter or click the button to apply

Apply Instruction

### Quick Improvements

**Make it more positive**  
Add optimism and warmth →

**Rewrite professionally**  
Polish for business context →

### Improvement Goals

Choose how you want to transform your text

Select Improvement Goal

**Increase Openness**  
Make the text more creative and open-minded

**Decrease Openness**  
Make the text more focused and conventional

**Reduce Aggressiveness**  
Soften aggressive or confrontational language

**Reduce Sadness**  
Transform melancholic tone to hopeful

**Make More Positive**  
Increase overall positivity and optimism

**Make Professional**  
Polish for business/formal contexts

# History

InnerVoice

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Settings

Manage your account settings and preferences

Profile

Appearance

Notifications

Security

Data & Privacy

Profile Information

Update your personal details

R

Change Avatar

JPG, PNG, GIF or WebP. Max size 2MB.

Username

Rishad1453

Email

rishad1453@gmail.com

Save Changes

R

Rishad1453

rishad1453@gmail.com

# Extension



**InnerVoice AI**  
Understand your emotions better

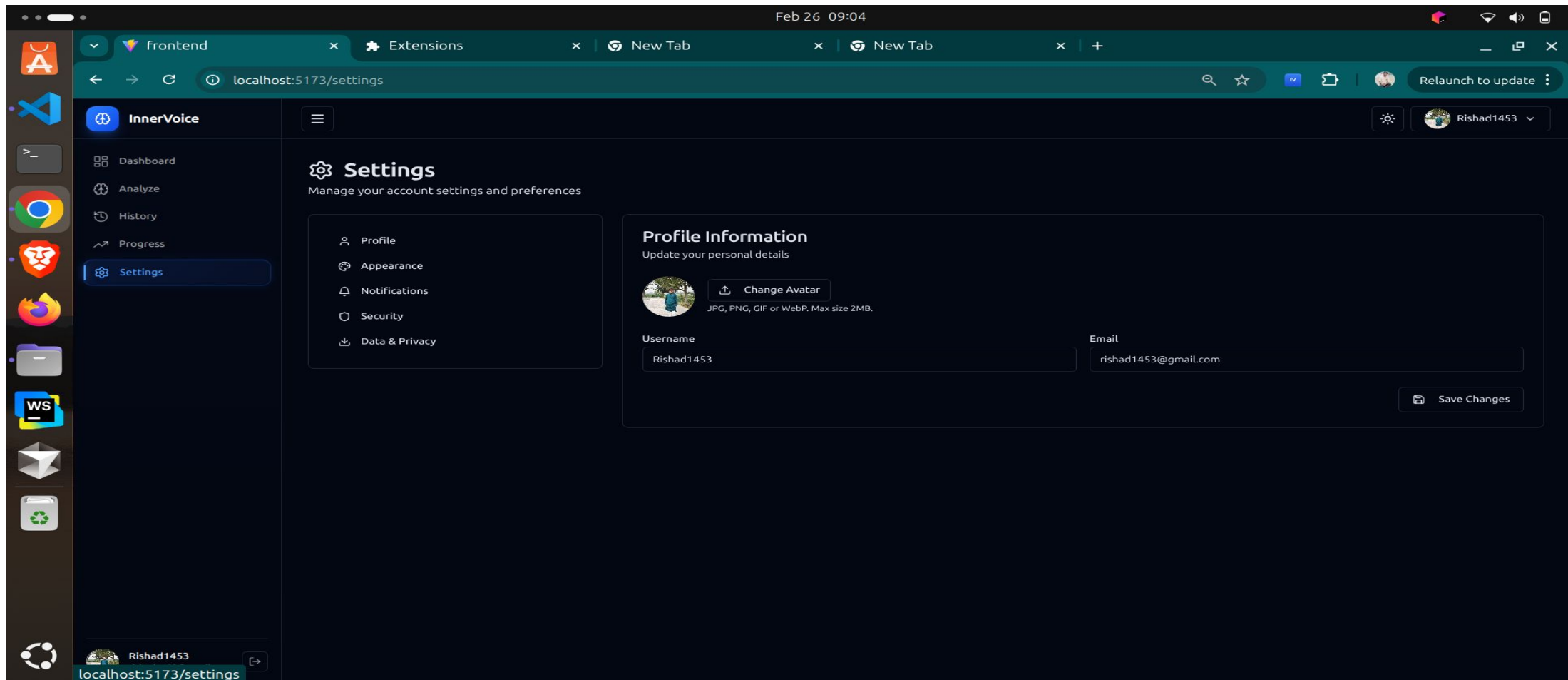
Login Register

**Username**

**Password**

Login

# Settings



# Tools

**Languages or Tools to be used:** Python, Typescript, React, Django, VS Code, GitHub, Colab, Virtual GPU

# Timeline

| <b>Weeks</b>         | <b>1st</b> | <b>2nd</b> | <b>3rd</b> | <b>4th</b> | <b>5th</b> | <b>6th</b> | <b>7th</b> | <b>8th</b> | <b>9th</b> | <b>10th</b> | <b>11th</b> | <b>12th</b> | <b>13th</b> | <b>14th</b> | <b>15th</b> | <b>16th</b> |
|----------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Proposal Writing     |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |             |
| SRS Documentation    |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |             |
| Development          |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |             |
| Testing              |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |             |
| Deployment           |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |             |
| Final report writing |            |            |            |            |            |            |            |            |            |             |             |             |             |             |             |             |

Thank You